

Statistical and Computational Models

Lecture 6: Maximum likelihood

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Roadmap

The likelihood

Finding the maximum

Two choices, revisited

Thus far, our approach to estimation has involved two choices:

Function class: The type of functions that we are allowed to fit (e.g. linear, polynomials, non-parametric).

Loss: What kind of 'error' matters, and how we price those errors.

Both of these are **choices** we explicitly make.

What if we turn this on its head, and assume **only** a probability model for $\mathbf{Y}$:

$$Y_i \sim p(y \mid \theta)$$

Maximum likelihood

Maximum likelihood approaches point estimation in a general way:

1. **Assume** the outcome Y_i is generated by a distribution that depends on some parameters (that we care about).
2. **Guess** values of the parameters of interest, and see how likely the observed data would be had those been the true parameters.
3. **Learn** the values of the parameters that maximize the likelihood of the observed data.

That is, we learn the values of our parameters that are mostly likely **given the data we observe** and the **probability model we assume**.

The likelihood

Assume our data $\mathbf{Y} = \{Y_1, \dots, Y_n\}$ came from a joint distribution (the DGP) with parameter θ : $\mathbf{Y} \sim p(\mathbf{Y} \mid \theta)$.

$p(\mathbf{Y} \mid \theta)$: the probability that we observe data $\mathbf{Y}$, given parameter θ .

But really $\mathbf{Y}$ is **known**, while θ is **unknown**, so we read $p(\mathbf{Y} \mid \theta)$ as a function of θ .

We call this the **likelihood function**:

$$L(\theta; \mathbf{Y}) = p(\mathbf{Y} \mid \theta)$$

The same thing as $p(\mathbf{Y} \mid \theta)$, but arguments reversed: the data are fixed, θ varies.

Building the likelihood function: Democracy and autocracy

Suppose we measure the regime type for $i = 1, \dots, 205$ countries in 2026: $Y_i = 1$ if a democracy, $Y_i = 0$ if an autocracy.

Let's assume that countries are democracies **with probability π** .

Our goal is to estimate π using our $n = 205$ **independent** country observations.

Let's start with just one country:

- $\mathbb{P}(Y_i = 1)? \pi$.
- $\mathbb{P}(Y_i = 0)? 1 - \pi$
- $\mathbb{P}(Y_i = y)? \pi^y(1 - \pi)^{1-y}$

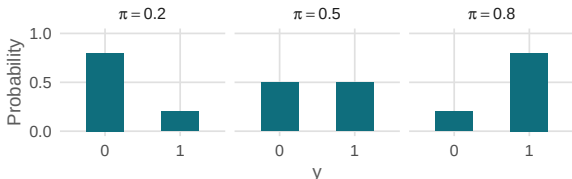
So, $Y_i \sim \text{Bernoulli}(\pi)$.

Brief aside: Bernoulli distribution

A **Bernoulli** distribution is the distribution of a single binary draw (one country, democracy or not), with one parameter π :

$$p(y \mid \pi) = \pi^y(1 - \pi)^{1-y}, \quad y \in \{0, 1\}$$

Read: the exponents switch on either π or $1 - \pi$.



$\mathbb{E}[Y_i] = \pi$ and $\text{Var}[Y_i] = \pi(1 - \pi)$, largest at $\pi = 0.5$.

Add up n independent Bernoulli draws and you get a Binomial(n, π).

Building the likelihood function: Democracy and autocracy

With two independent countries, the joint probability is the product:

$$\begin{aligned}\mathbb{P}(Y_1 = y_1, Y_2 = y_2) &= \mathbb{P}(Y_1 = y_1) \mathbb{P}(Y_2 = y_2) \\ &= \left[\pi^{y_1} (1 - \pi)^{1-y_1} \right] \left[\pi^{y_2} (1 - \pi)^{1-y_2} \right] \\ &= \pi^{y_1+y_2} (1 - \pi)^{(1-y_1)+(1-y_2)}\end{aligned}$$

Note: This operation works because we assumed **independence**.

Building the likelihood function: Democracy and autocracy

... with all n countries:

$$\begin{aligned}\mathbb{P}(Y_1 = y_1, \dots, Y_n = y_n) &= \prod_{i=1}^n \mathbb{P}(Y_i = y_i) = \prod_{i=1}^n \pi^{y_i} (1 - \pi)^{1-y_i} \\ &= \pi^r (1 - \pi)^{n-r} \quad (\text{for } r \text{ democracies})\end{aligned}$$

We can call this quantity:

- the 'probability of the data' given π
- $p(\mathbf{Y} \mid \pi)$
- the **sample likelihood**, $L(\pi; \mathbf{Y})$

But L is a function of π , **not** a distribution for it.

In our example, $r = 121$ of $n = 205$. Try to guess π ...

Guessing π

Let's try five guesses for π , and calculate the likelihood given our 205 observed outcomes:

π	$L(\pi; \mathbf{Y})$	$\ell(\pi)$
0.30	5.2e-77	-175.64
0.50	1.9e-62	-142.10
0.59	5.6e-61	-138.74
0.70	2.2e-63	-144.29
0.90	2.9e-90	-206.17

The third row is r/n . This π maximizes L (given the search space). But note how every L is tiny because they are products of 205 numbers, each below one.

We thus typically take the **log**: the product of n factors becomes a sum of n terms.

The last column gives this **log-likelihood**, $\ell(\pi) = \log L$. The log is monotone with respect to the likelihood, so it preserves order and the maximum does not change.

Maximum likelihood estimation

Recall our **estimation goal**: make our best guess at θ , given the data we observe.

$L(\hat{\theta}; \mathbf{Y})$ tells us how likely it is that we observe our data, if $\theta = \hat{\theta}$. Our optimisation task is to find the $\hat{\theta}$ that makes L the largest – the **maximum likelihood estimator** (MLE).

Working with the **log-likelihood**:

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta; \mathbf{Y}) = \arg \max_{\theta} \ell(\theta), \quad \ell(\theta) = \log L(\theta; \mathbf{Y})$$

Consistency: if the model is true, we can show that $\hat{\theta}_{\text{MLE}} \xrightarrow{P} \theta$.

The MLE for our countries

So, to find the MLE for our countries:

The likelihood, and the log-likelihood:

$$L(\pi; \mathbf{Y}) = \prod_{i=1}^n \pi^{Y_i} (1 - \pi)^{1 - Y_i} = \pi^r (1 - \pi)^{n-r}$$
$$\ell(\pi) = r \log \pi + (n - r) \log(1 - \pi)$$

The first derivative of the log-likelihood is called the **score**:

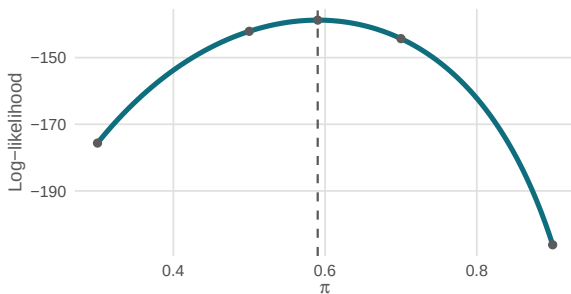
$$s(\pi) = \frac{\partial}{\partial \pi} \ell(\pi) = \frac{r}{\pi} - \frac{n - r}{1 - \pi}$$

Solving $s(\hat{\pi}) = 0$, we get $\hat{\pi}_{\text{MLE}} = r/n = 0.590$.

The MLE for the population proportion is the sample proportion. We've actually seen this before... The intercept-only OLS model.

The log-likelihood

$$\ell(\pi) = r \log \pi + (n - r) \log(1 - \pi), \quad r = \sum_{i=1}^n y_i$$



The grey dots here show the five guesses of π and their ℓ .

The loss you were using

For any model, we maximise a sum over countries:

$$\ell(\theta) = \sum_{i=1}^n \ell_i(\theta), \quad \ell_i(\theta) = \log p(y_i | \theta)$$

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \ell(\theta) = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \{ -\log p(y_i | \theta) \}$$

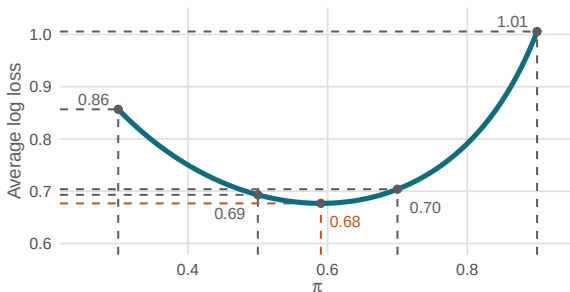
Read: the second line is just a **training loss**, like $\text{MSE}_{\text{train}}$. A democracy adds $-\log \pi$, an autocracy $-\log(1 - \pi)$.

At $\hat{\pi} = 0.590$: each democracy adds $-\log \hat{\pi} = 0.53$, each autocracy $-\log(1 - \hat{\pi}) = 0.89$. The average over our 205 countries is 0.68.

But here the **log loss** comes with $p(y | \theta)$ – we did not **choose** it.

The log loss for every π

$$\frac{1}{n} \sum_{i=1}^n \left\{ -\log p(y_i | \pi) \right\} = -\frac{\ell(\pi)}{n}$$



The same curve as the log-likelihood, upside down. For any other π the loss is higher than $\hat{\pi}_{\text{MLE}} = 0.590$.

The loss you were using

Now consider a normal Y_i , with covariates:

$$Y_i \sim N(\mathbf{x}_i^\top \boldsymbol{\beta}, \sigma^2) \Rightarrow -\ell(\mathbf{b}) = \frac{\text{RSS}(\mathbf{b})}{2\sigma^2} + \frac{n}{2} \log(2\pi\sigma^2)$$

Only RSS depends on $\mathbf{b}$ (that π is 3.14...).

So if Y_i is normal, lecture 2's least squares **is** maximum likelihood!

Two choices, revisited

So far we have made just one assumption, but that assumption gives us both the function class and the loss:

Assume	The class	The loss
$Y_i \sim \text{Bernoulli}(\pi)$	$\{\mathbf{x} \mapsto \pi\}$, constant	$-\log p(y \mid \pi)$
$Y_i \sim N(\mathbf{x}_i^\top \beta, \sigma^2)$	$\{\mathbf{x} \mapsto \mathbf{x}^\top \beta\}$, linear	$(y - \mathbf{x}^\top \beta)^2$

Roadmap

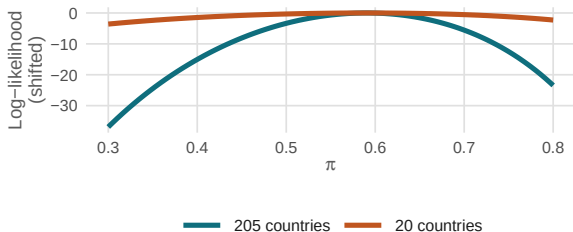
The likelihood

Finding the maximum

Fisher information

The score is the slope of ℓ . Its **curvature** is the second derivative, the **Hessian**: $H(\theta) = \partial^2 \ell / \partial \theta^2$.

Which of these log-likelihoods contains more **information** about π ?



— H large implies that ℓ is spiky and carries much information, giving a precise location of θ .

Climbing the likelihood

Usually $\mathbf{s}(\theta) = \mathbf{0}$ has no closed-form solution (e.g. if we include covariates, as in the next lecture).

Naive approach: a **grid search**, as in the log-likelihood curve. Infeasible unless θ has few dimensions, so instead we use an algorithm.

Suppose you are at $\theta^{(t)}$ – where should you move next?

Slope $\mathbf{s}(\theta)$ positive, move right; negative, move left. Curvature $\mathbf{H}(\theta)$ steep, move a little; flat, move a lot:

$$\theta^{(t+1)} = \theta^{(t)} - \mathbf{H}(\theta^{(t)})^{-1} \mathbf{s}(\theta^{(t)})$$

This is the **Newton–Raphson** algorithm.

Climbing the likelihood

On our log-likelihood, from a bad start, $\pi^{(0)} = 0.9$:

t	$\pi^{(t)}$	$s(\pi^{(t)})$	$\ell(\pi^{(t)})$
0	0.9000	-705.6	-206.17
1	0.8175	-312.2	-167.26
2	0.7019	-109.5	-144.50
3	0.6101	-17.1	-138.91
4	0.5906	-0.3	-138.74
5	0.5902	0.0	-138.74
6	0.5902	0.0	-138.74

After five steps Newton–Raphson finds a **zero** of the score.

`glm()` uses the expected Hessian instead (**Fisher scoring**).

optim()

In R we call this through `optim()`, from the same start:

```
loglik <- function(p) r * log(p) + (n - r) * log(1 - p)

opt <- optim(par = 0.9, fn = loglik, method = "BFGS",
            control = list(fnscale = -1), hessian = TRUE)
```

`optim()` minimises by default; `fnscale = -1` flips the objective. BFGS is **quasi-Newton**: Newton's update, with H approximated.

`hessian = TRUE` returns the numerical Hessian at the maximum.

`optim()`: 0.5902; closed form: 0.5902.

References

- Efron, B. & Hastie, T. (2016) *Computer Age Statistical Inference* – **4.1–4.2**.
- Rodríguez, G. *Generalized Linear Models*, Princeton lecture notes – **Appendix A**.
grodriguez.github.io/glms.